



**Laboratorium
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Laporan Akhir Praktikum Jaringan Komputer

VLAN - TRUNK - OSPF - MULTIVENDOR

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2026

1 Langkah-Langkah Percobaan

1.1 Praktikum 1: Konfigurasi Cisco Switch

1. Masuk ke terminal Cisco Switch menggunakan perintah `enable` dilanjutkan dengan `configure terminal`.
2. Membuat VLAN 10 (Finance), VLAN 20 (HR), VLAN 30 (IT), dan VLAN 40 (Marketing) pada switch.
3. Melakukan verifikasi pembuatan VLAN dengan perintah `show vlan brief` untuk memastikan statusnya aktif.

VLAN	Name	Status	Ports
1	default	active	Gi0/0, Gi0/1, Gi0/2, Gi0/3 Gi1/0, Gi1/1, Gi1/2, Gi1/3
10	Finance	active	
20	HR	active	
30	IT	active	
40	Marketing	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

4. Mengonfigurasi *Access port* untuk setiap VLAN yang disesuaikan dengan interface pada Cisco Switch (Gi0/2 untuk VLAN 10, Gi0/1 untuk VLAN 20, Gi1/1 untuk VLAN 30, dan Gi0/3 untuk VLAN 40).
5. Mengonfigurasi port Gi0/0 sebagai *Trunk port* menuju Cisco Router dan mengizinkan VLAN 10, 20, 30, serta 40 untuk melewatinya.
6. Melakukan verifikasi konfigurasi trunk dengan perintah `show interfaces trunk` dan menyimpan konfigurasi.

```
Switch(config)#interface gi0/0
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#description TRUNK-TO-CISCO-ROUTER
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#exit
Switch#show interface
*Jun  6 01:26:21.533: %SYS-5-CONFIG_I: Configured from console by console trunk

Port      Mode           Encapsulation  Status        Native vlan
Gi0/0     on             802.1q         trunking      1

Port      Vlans allowed on trunk
Gi0/0     10,20,30,40

Port      Vlans allowed and active in management domain
Gi0/0     10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     none
Switch#
```

1.2 Praktikum 2: Konfigurasi Cisco Router

1. Masuk ke Cisco Router dan mengonfigurasi interface Gi0/2 (yang terhubung ke switch) agar menyala tanpa IP address.

2. Membuat *subinterface* (ROAS) pada Gi0/2 untuk VLAN 10, 20, 30, dan 40 beserta IP *gateway* masing-masing (192.168.10.1, 192.168.20.1, 192.168.30.1, 192.168.40.1) dengan enkapsulasi dot1Q.
3. Mengonfigurasi DHCP-Server pada router khusus untuk VLAN 10 dan VLAN 20, serta mengecualikan 10 IP pertama.
4. Mengonfigurasi IP Static secara manual pada VPCS untuk VLAN 30 dan VLAN 40.
5. Memverifikasi interface router dengan perintah `show ip interface brief`.

```
Router#show ip interface brief
Interface          IP-Address      OK? Method Status        Protocol
GigabitEthernet0/0 unassigned      YES unset  administratively down down
GigabitEthernet0/1 unassigned      YES unset  administratively down down
GigabitEthernet0/2 unassigned      YES unset  up            up
GigabitEthernet0/2.10 192.168.10.1   YES manual up            up
GigabitEthernet0/2.20 192.168.20.1   YES manual up            up
GigabitEthernet0/2.30 192.168.30.1   YES manual up            up
GigabitEthernet0/2.40 192.168.40.1   YES manual up            up
GigabitEthernet0/3 unassigned      YES unset  administratively down down
```

6. Mengecek DHCP pool yang telah dibuat dengan perintah `show ip dhcp pool`.

```
Router#show ip dhcp pool
Pool VLAN10-FINANCE :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 254
  Leased addresses                : 1
  Pending event                  : none
  1 subnet is currently in the pool :
    Current index  IP address range  Leased addresses
    192.168.10.12  192.168.10.1 - 192.168.10.254  1
Pool VLAN20-HR :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 254
  Leased addresses                : 1
  Pending event                  : none
  1 subnet is currently in the pool :
    Current index  IP address range  Leased addresses
    192.168.20.3   192.168.20.1 - 192.168.20.254  1
```

7. Meminta IP DHCP pada VPCS VLAN 10 dan 20, lalu memverifikasinya.

```
VPCS> ip dhcp
DDORA IP 192.168.10.11/24 GW 192.168.10.1
```

```
VPCS> ip dhcp
DDORA IP 192.168.20.2/24 GW 192.168.20.1
```

8. Melakukan uji ping ke masing-masing *gateway* dari setiap PC VLAN untuk memastikan konektivitas.

```
VPCS> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=3.255 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=3.640 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=3.550 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=3.891 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=3.657 ms
```

```
VPCS> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=3.310 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=3.520 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=3.233 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=3.964 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=3.214 ms
```

```
VPCS> ping 192.168.30.1

84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=5.552 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=4.189 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=4.392 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=3.669 ms
84 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=5.868 ms
```

```
VPCS> ping 192.168.40.1

84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=4.096 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=3.859 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=3.369 ms
84 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=5.501 ms
84 bytes from 192.168.40.1 icmp_seq=5 ttl=255 time=2.756 ms
```

9. Melakukan uji ping antar-VLAN untuk memastikan fungsi *routing* inter-VLAN berjalan.

```
VPCS> ping 192.168.20.2

84 bytes from 192.168.20.2 icmp_seq=1 ttl=63 time=8.742 ms
84 bytes from 192.168.20.2 icmp_seq=2 ttl=63 time=5.633 ms
84 bytes from 192.168.20.2 icmp_seq=3 ttl=63 time=7.606 ms
84 bytes from 192.168.20.2 icmp_seq=4 ttl=63 time=5.934 ms
84 bytes from 192.168.20.2 icmp_seq=5 ttl=63 time=4.419 ms
```

```

VPCS> ping 192.168.30.10

64 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=4.147 ms
64 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=5.590 ms
64 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=4.467 ms
64 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=6.627 ms
64 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=4.885 ms

VPCS> ping 192.168.40.10

64 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=4.385 ms
64 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=5.441 ms
64 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=4.570 ms
64 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=5.438 ms
64 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=4.582 ms

```

10. Mengecek klien yang mendapatkan IP dari DHCP dengan perintah `show ip dhcp binding` di Cisco Router.

```

Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/          Lease expiration        Type
                   Hardware address/
                   User name
192.168.10.11       0100.5079.6668.3e   Jun 07 2026 01:37 AM    Automatic
192.168.20.2        0100.5079.6668.3d   Jun 07 2026 01:37 AM    Automatic

```

1.3 Praktikum 3: Konfigurasi Huawei Router dan IP Address Antarrouter

1. Mengonfigurasi IP address untuk LAN Huawei (192.168.50.1/24) pada interface Ethernet1/0/2.

```

<HUAWEI>system-view
Enter system view, return user view with return command.
[~HUAWEI]interface Ethernet1/0/2
[~HUAWEI-Ethernet1/0/2]description LAN-HUAWEI
[*HUAWEI-Ethernet1/0/2]ip address 192.168.50.1 255.255.255.0
[*HUAWEI-Ethernet1/0/2]undo shutdown
[*HUAWEI-Ethernet1/0/2]quit
[*HUAWEI]commit
[~HUAWEI]

```

2. Mengatur IP static pada VPC LAN Huawei dan melakukan ping ke *gateway* Huawei.

```

VPCS> ip 192.168.50.10/24 192.168.50.1
Checking for duplicate address...
PC1 : 192.168.50.10 255.255.255.0 gateway 192.168.50.1

VPCS> show ip

NAME       : VPCS[1]
IP/MASK     : 192.168.50.10/24
GATEWAY     : 192.168.50.1
DNS         :
MAC         : 00:50:79:66:68:3f
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU         : 1500

VPCS> █

```

```

VPCS> ping 192.168.50.1

84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=9.636 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=3.653 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=3.067 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=255 time=3.769 ms
84 bytes from 192.168.50.1 icmp_seq=5 ttl=255 time=2.662 ms

```

3. Mengonfigurasi IP address untuk Link 1 antara Cisco Router (Gi0/3) dan Huawei Router (Eth1/0/0) menggunakan network 10.10.10.0/30.
4. Mengonfigurasi IP address untuk Link 2 antara Cisco Router (Gi0/0) dan Huawei Router (Eth1/0/4) menggunakan network 10.10.11.0/30.
5. Mengonfigurasi IP address untuk link antara Huawei Router (Eth1/0/3) dan MikroTik (ether1) menggunakan network 10.10.20.0/30.
6. Mengonfigurasi IP address untuk link antara Cisco Router (Gi0/1) dan MikroTik (ether2) menggunakan network 10.10.30.0/30.
7. Memverifikasi IP di Cisco Router dengan show ip interface brief.

```

Router#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	10.10.11.2	YES	manual	up	up
GigabitEthernet0/1	10.10.30.1	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	unset	up	up
GigabitEthernet0/2.10	192.168.10.1	YES	manual	up	up
GigabitEthernet0/2.20	192.168.20.1	YES	manual	up	up
GigabitEthernet0/2.30	192.168.30.1	YES	manual	up	up
GigabitEthernet0/2.40	192.168.40.1	YES	manual	up	up
GigabitEthernet0/3	10.10.10.1	YES	manual	up	up

```

Router#

```

8. Memverifikasi IP di Huawei Router dengan display ip interface brief.

Interface	IP Address/Mask	Physical	Protocol	VPN
Ethernet1/0/0	10.10.10.2/30	up	up	--
Ethernet1/0/0.4094	128.1.138.137/16	up	up	l3vpn
Ethernet1/0/1	unassigned	up	down	--
Ethernet1/0/1.4094	128.1.138.137/16	up	up	l3vpn
Ethernet1/0/2	192.168.50.1/24	up	up	--
Ethernet1/0/2.4094	128.1.138.137/16	up	up	l3vpn
Ethernet1/0/3	10.10.20.1/30	up	up	--
Ethernet1/0/3.4094	128.1.138.137/16	up	up	l3vpn
Ethernet1/0/4	10.10.11.2/30	up	up	--
Ethernet1/0/4.4094	128.1.138.137/16	up	up	l3vpn

9. Memverifikasi IP di MikroTik dengan perintah `/ip address print`.

```
Flags: X - disabled, I - invalid, D - dynamic
```

#	ADDRESS	NETWORK	INTERFACE
0	;;; TO-HUAWEI 10.10.20.2/30	10.10.20.0	ether1
1	;;; TO-CISCO 10.10.30.2/30	10.10.30.0	ether2

10. Menguji koneksi ping antar router (Cisco ke Huawei dan MikroTik, Huawei ke Cisco dan MikroTik, serta MikroTik ke Cisco dan Huawei).

```
Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/2/4 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
..!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
```

```
[~HUAWEI]ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL_C to break
  Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=2 ms
  Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=2 ms
  Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=2 ms

--- 10.10.20.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 1/1/2 ms

[~HUAWEI]ping 10.10.10.1
PING 10.10.10.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=3 ms
  Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=3 ms
  Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=3 ms
  Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=3 ms
  Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=3 ms
```

```
[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=2 ms

--- 10.10.11.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 2/2/3 ms
```

```
[admin@MikroTik] > ping 10.10.20.1
SEQ HOST                                SIZE TTL TIME  STATUS
  0 10.10.20.1                          56 255 4ms
  1 10.10.20.1                          56 255 1ms
  2 10.10.20.1                          56 255 1ms
  3 10.10.20.1                          56 255 1ms
  4 10.10.20.1                          56 255 1ms
  5 10.10.20.1                          56 255 1ms
  6 10.10.20.1                          56 255 1ms
sent=7 received=7 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=4ms

[admin@MikroTik] > ping 10.10.30.1
SEQ HOST                                SIZE TTL TIME  STATUS
  0 10.10.30.1                          56 255 1ms
  1 10.10.30.1                          56 255 2ms
  2 10.10.30.1                          56 255 1ms
  3 10.10.30.1                          56 255 2ms
  4 10.10.30.1                          56 255 1ms
  5 10.10.30.1                          56 255 1ms
  6 10.10.30.1                          56 255 1ms
sent=7 received=7 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=2ms
```

1.4 Praktikum 4: Konfigurasi OSPF Multivendor dan OSPF Cost

1. Mengonfigurasi OSPF pada Cisco Router dengan *router-id* 1.1.1.1, mendaftarkan seluruh *network* yang terhubung, dan mengatur *cost* untuk Link 1, Link 2, dan jalur *backup* ke MikroTik.
2. Memverifikasi ketersediaan *network* OSPF pada Cisco dengan `show ip protocols`.

```
Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.255 area 0
    192.168.20.0 0.0.0.255 area 0
    192.168.30.0 0.0.0.255 area 0
    192.168.40.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 110)
```

3. Mengonfigurasi OSPF pada Huawei Router dengan *router-id* 2.2.2.2, mendaftarkan *network*, dan mengatur OSPF *cost* sesuai ketentuan topologi.

- Memverifikasi OSPF *peer* Huawei ke Cisco dan mengecek *routing table* OSPF.

```
Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3 Address: 10.10.20.2
  State: Full Mode:Nbr is Master Priority: 1
  DR: 10.10.20.1 BDR: 10.10.20.2 MTU: 1500
  Dead timer due in 34 sec
  Retrans timer interval: 5
  Neighbor is up for 00h01m46s
  Neighbor Up Time : 2026-06-06 02:45:07
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.11.1
  State: Full Mode:Nbr is Slave Priority: 1
  DR: 10.10.11.1 BDR: 10.10.11.2 MTU: 1500
  Dead timer due in 34 sec
  Retrans timer interval: 5
  Neighbor is up for 00h09m20s
  Neighbor Up Time : 2026-06-06 02:37:34
  Authentication Sequence: [ 0 ]
```

```
Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

 10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O   10.10.20.0/30 [110/110] via 10.10.11.2, 00:11:06, GigabitEthernet0/0
    [110/110] via 10.10.10.2, 00:28:08, GigabitEthernet0/3
O   192.168.50.0/24 [110/11] via 10.10.11.2, 00:11:06, GigabitEthernet0/0
    [110/11] via 10.10.10.2, 00:29:13, GigabitEthernet0/3
```

```
OSPF Process 1 with Router ID 2.2.2.2
Routing Tables

Routing for Network
Destination      Cost      Type      NextHop      AdvRouter      Area
10.10.10.0/30    10        Direct    10.10.10.2    2.2.2.2        0.0.0.0
10.10.11.0/30    10        Direct    10.10.11.2    2.2.2.2        0.0.0.0
10.10.20.0/30    100       Direct    10.10.20.1    2.2.2.2        0.0.0.0
10.10.30.0/30    110       Stub      10.10.11.1    1.1.1.1        0.0.0.0
10.10.30.0/30    110       Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.10.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.10.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.20.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.20.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.30.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.30.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.40.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.40.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.50.0/24  1         Direct    192.168.50.1  2.2.2.2        0.0.0.0
```

- Mengonfigurasi MikroTik sebagai jalur *backup* dengan *router-id* 3.3.3.3, menambahkan *network* ke *backbone area*, dan mengatur *cost* 100 pada kedua interfacenya.
- Memverifikasi *neighbor* dan *routing table* OSPF di MikroTik.

```
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m59s

1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m55s
```

```
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADo 0.0.0.0/0 10.10.30.1 110
1 ADo 10.10.10.0/30 10.10.20.1 110
2 ADo 10.10.11.0/30 10.10.30.1 110
3 ADo 192.168.10.0/24 10.10.30.1 110
4 ADo 192.168.20.0/24 10.10.30.1 110
5 ADo 192.168.30.0/24 10.10.30.1 110
6 ADo 192.168.40.0/24 10.10.30.1 110
7 ADo 192.168.50.0/24 10.10.20.1 110
```

7. Memverifikasi kembali status *neighbor* secara menyeluruh di Cisco Router dan Huawei Router.

```
Router#show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
2.2.2.2 1 FULL/BDR 00:00:32 10.10.11.2 GigabitEthernet0/0
3.3.3.3 1 FULL/BDR 00:00:31 10.10.30.2 GigabitEthernet0/1
2.2.2.2 1 FULL/BDR 00:00:34 10.10.10.2 GigabitEthernet0/3
```

```
Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3 Address: 10.10.20.2
State: Full Mode:Nbr is Master Priority: 1
DR: 10.10.20.1 BDR: 10.10.20.2 MTU: 1500
Dead timer due in 34 sec
Retrans timer interval: 5
Neighbor is up for 00h01m46s
Neighbor Up Time : 2026-06-06 02:45:07
Authentication Sequence: [ 0 ]
```

```
Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.11.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.11.1 BDR: 10.10.11.2 MTU: 1500
Dead timer due in 34 sec
Retrans timer interval: 5
Neighbor is up for 00h09m20s
Neighbor Up Time : 2026-06-06 02:37:34
Authentication Sequence: [ 0 ]
```

8. Mengecek *routing table* menuju LAN Huawei dari Cisco Router.

```
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
Known via "ospf 1", distance 110, metric 11, type intra area
Last update from 10.10.11.2 on GigabitEthernet0/0, 00:10:35 ago
Routing Descriptor Blocks:
10.10.11.2, from 2.2.2.2, 00:10:35 ago, via GigabitEthernet0/0
Route metric is 11, traffic share count is 1
* 10.10.10.2, from 2.2.2.2, 00:28:42 ago, via GigabitEthernet0/3
Route metric is 11, traffic share count is 1
```

9. Mengecek semua rute OSPF di Cisco Router, Huawei Router, dan MikroTik.

```
Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O    10.10.20.0/30 [110/110] via 10.10.11.2, 00:11:06, GigabitEthernet0/0
    [110/110] via 10.10.10.2, 00:28:08, GigabitEthernet0/3
O    192.168.50.0/24 [110/11] via 10.10.11.2, 00:11:06, GigabitEthernet0/0
    [110/11] via 10.10.10.2, 00:29:13, GigabitEthernet0/3
```

```
OSPF Process 1 with Router ID 2.2.2.2
Routing Tables

Routing for Network
Destination      Cost      Type      NextHop      AdvRouter      Area
10.10.10.0/30    10        Direct    10.10.10.2    2.2.2.2        0.0.0.0
10.10.11.0/30    10        Direct    10.10.11.2    2.2.2.2        0.0.0.0
10.10.20.0/30    100       Direct    10.10.20.1    2.2.2.2        0.0.0.0
10.10.30.0/30    110       Stub      10.10.11.1    1.1.1.1        0.0.0.0
10.10.30.0/30    110       Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.10.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.10.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.20.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.20.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.30.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.30.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.40.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.40.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.50.0/24  1         Direct    192.168.50.1  2.2.2.2        0.0.0.0
```

```
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit

#      DST-ADDRESS      PREF-SRC      GATEWAY      DISTANCE
0 ADo  0.0.0.0/0          10.10.30.1    110
1 ADo  10.10.10.0/30     10.10.20.1    110
      10.10.30.1
2 ADo  10.10.11.0/30     10.10.20.1    110
      10.10.30.1
3 ADo  192.168.10.0/24   10.10.30.1    110
4 ADo  192.168.20.0/24   10.10.30.1    110
5 ADo  192.168.30.0/24   10.10.30.1    110
6 ADo  192.168.40.0/24   10.10.30.1    110
7 ADo  192.168.50.0/24   10.10.20.1    110
```

10. Melakukan pengujian *ping end-to-end* dari setiap PC VLAN menuju LAN Huawei.

```
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=11.906 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=7.849 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=7.763 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=7.797 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=5.920 ms
```

```
VPCS> ping 192.168.50.10  
  
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=5.311 ms  
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=5.882 ms  
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=5.913 ms  
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=7.667 ms  
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=6.199 ms
```

```
VPCS> ping 192.168.50.10  
  
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=5.592 ms  
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=6.095 ms  
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=5.360 ms  
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=6.104 ms  
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=6.253 ms
```

```
VPCS> ping 192.168.50.10  
  
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=6.139 ms  
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=5.861 ms  
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=6.204 ms  
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=5.708 ms  
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=8.263 ms
```

11. Melakukan pengujian ping balik dari PC LAN Huawei ke semua network VLAN Cisco.

```
4 bytes from 192.168.10.11 icmp_seq=1 ttl=62 time=9.932 ms  
4 bytes from 192.168.10.11 icmp_seq=2 ttl=62 time=7.668 ms  
4 bytes from 192.168.10.11 icmp_seq=3 ttl=62 time=6.977 ms  
4 bytes from 192.168.10.11 icmp_seq=4 ttl=62 time=5.721 ms  
4 bytes from 192.168.10.11 icmp_seq=5 ttl=62 time=7.916 ms
```

```
PCS> ping 192.168.20.11  
  
92.168.20.11 icmp_seq=1 timeout  
92.168.20.11 icmp_seq=2 timeout  
92.168.20.11 icmp_seq=3 timeout  
C  
PCS> ping 192.168.20.2
```

```
4 bytes from 192.168.20.2 icmp_seq=1 ttl=62 time=9.844 ms  
4 bytes from 192.168.20.2 icmp_seq=2 ttl=62 time=8.476 ms  
4 bytes from 192.168.20.2 icmp_seq=3 ttl=62 time=5.932 ms  
4 bytes from 192.168.20.2 icmp_seq=4 ttl=62 time=6.758 ms  
4 bytes from 192.168.20.2 icmp_seq=5 ttl=62 time=5.967 ms
```

```

VPCS> ping 192.168.30.10

64 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=8.537 ms
64 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=7.069 ms
64 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=6.146 ms
64 bytes from 192.168.30.10 icmp_seq=4 ttl=62 time=6.906 ms
64 bytes from 192.168.30.10 icmp_seq=5 ttl=62 time=7.486 ms

VPCS> ping 192.168.40.10

64 bytes from 192.168.40.10 icmp_seq=1 ttl=62 time=7.399 ms
64 bytes from 192.168.40.10 icmp_seq=2 ttl=62 time=6.196 ms
64 bytes from 192.168.40.10 icmp_seq=3 ttl=62 time=7.663 ms
64 bytes from 192.168.40.10 icmp_seq=4 ttl=62 time=7.585 ms
64 bytes from 192.168.40.10 icmp_seq=5 ttl=62 time=6.206 ms

```

1.5 Praktikum 5: Pengujian Failover OSPF

1. Melakukan *trace route* dari PC VLAN menuju LAN Huawei (192.168.50.10) dalam kondisi normal, di mana kedua *link* utama aktif.

```

Router#trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
VRF info: (vrf in name/id, vrf out name/id)
 0 10.10.10.2 8 msec
 1 10.10.11.2 3 msec
 2 10.10.10.2 3 msec
 3 192.168.50.10 9 msec 6 msec 4 msec

```

2. Mematikan antarmuka Gi0/3 pada Cisco Router (1 *link* Cisco-Huawei mati) dan melakukan *trace route* kembali untuk membuktikan bahwa *traffic* tetap melewati jalur utama yang tersisa.

```

Router#show ip route 192.168.50.10
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:37:20 ago
  Routing Descriptor Blocks:
    * 10.10.11.2, from 2.2.2.2, 00:37:20 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
Router#trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
VRF info: (vrf in name/id, vrf out name/id)
 0 10.10.11.2 3 msec 2 msec 2 msec
 1 192.168.50.10 6 msec 4 msec 5 msec
Router#

```

3. Mematikan antarmuka Gi0/0 pada Cisco Router (kedua *link* utama mati), mengecek pergerakan *routing*, lalu melakukan *trace route* untuk membuktikan *traffic* berpindah melalui jalur *backup* (MikroTik).

```

Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 201, type intra area
  Last update from 10.10.30.2 on GigabitEthernet0/1, 00:00:28 ago
  Routing Descriptor Blocks:
    * 10.10.30.2, from 2.2.2.2, 00:00:28 ago, via GigabitEthernet0/1
      Route metric is 201, traffic share count is 1
Router#trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
VRF info: (vrf in name/id, vrf out name/id)
  1 10.10.30.2 5 msec 1 msec 3 msec
  2 10.10.20.1 5 msec 3 msec 2 msec
  3 192.168.50.10 11 msec 5 msec 5 msec
Router#

```

4. Menghidupkan kembali antarmuka Gi0/0 dan Gi0/3 pada Cisco Router, lalu mengecek *trace route* yang seharusnya menunjukkan *routing* telah kembali ke jalur utama Cisco-Huawei.

```

Router#trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
VRF info: (vrf in name/id, vrf out name/id)
  1 10.10.10.2 8 msec
    10.10.11.2 3 msec
    10.10.10.2 3 msec
  2 192.168.50.10 9 msec 6 msec 4 msec

```